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Exploration of the oxidation chemistry of dimethoxymethane: Jet-stirred reactor experiments and kinetic modeling

Wenyu Sun[a,b], Tao Tao[a,b], Maxence Lailliau[c,d], Nils Hansen[e], Bin Yang[a,b,*], Philippe Dagaut[c,**]

[a] Center for Combustion Energy and Department of Energy and Power Engineering, Tsinghua University, Beijing 100084, PR China
[b] Key Laboratory for Thermal Science and Power Engineering of MOE, Tsinghua University, Beijing 100084, PR China
[c] CNRS-INSIS, I.C.A.R.E.,1C, Avenue de la recherche scientifique, 45071 Orléans Cedex 2, France
[d] Université d'Orléans, 6 Avenue du Parc Floral, 45100 Orléans, France
[e] Combustion Research Facility, Sandia National Laboratories, Livermore, CA 94551, USA

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ABSTRACT

Dimethoxymethane (DMM, $CH_3OCH_2OCH_3$) is a practical diesel additive, as well as a simple homologue in the class of polyoxymethylene dimethyl ethers (POMDMEs) which are considered as promising alternative fuels. To acquire an in-depth knowledge of DMM oxidation kinetics, the chemistry of an externally heated fuel-lean ($\phi = 0.5$) DMM/O_2/Ar mixture was investigated in a jet-stirred reactor (JSR) operated at near-atmospheric pressure (750 Torr). A molecular-beam mass spectrometer (MBMS) employing synchrotron photoionization was used to probe reactive intermediates. High-pressure (10 atm) oxidation experiments covering different equivalence ratios (0.2, 0.5 and 1.5) were carried out using another JSR facility equipped with gas chromatography (GC) and Fourier transform infrared spectrometry (FTIR) for speciation measurements. A new kinetic model was constructed and validated against the current measurements as well as those reported in literature. No obvious low-temperature reactivity was observed for DMM under the investigated conditions, though DMM has a long enough chain to allow internal hydrogen transfers leading to chain-branching. The kinetic modeling showed that hydrogen abstractions from the central (–OCH_2O–) moiety are favored, producing dominantly the $CH_3O\dot{C}HOCH_3$ fuel radical, which then rapidly decomposes instead of leading to chain-branching via O_2 addition. In contrast, the minor fuel radical $CH_3OCH_2O\dot{C}H_2$ can go through the O_2 addition and the subsequent isomerization steps, as confirmed by the detection of cyclic ether species. Another impact of the fast CH_3 production from $CH_3O\dot{C}HOCH_3$ β-scission is that CH_3O_2H serves as an important OH provider, facilitating the fuel consumption at medium temperatures. Major fuel destruction patterns could also apply to larger POMDME compounds.

1. Introduction

Polyoxymethylene dimethyl ethers (POMDMEs), with the general formula of $CH_3O(CH_2O)_nCH_3$, are appealing alternative fuels [1] which can inhibit the formation of multiple pollutants by breaking the soot-NO_x trade-off [2–4]. Dimethyl ether (DME, CH_3OCH_3) and dimethoxymethane (DMM, $CH_3OCH_2OCH_3$) can be seen as particular cases respectively with $n = 0$ and 1 in the POMDME family. Both individual compounds have been applied as clean diesel additives though they are not considered among the optimal POMDME fuel mixture with 3–4 (–CH_2O–) units [5]. It is interesting to note that DMM has a much lower cetane number (CN = 29) than all other compounds in the POMDME family (see Fig. S1) [2], even including DME, such a short-chain molecule (CN = 55). Cetane numbers can reflect fuel ignition properties which are governed by the low-temperature oxidation kinetics.

For DME and DMM, investigations into their oxidation kinetics are required for their controllable applications in engines. Furthermore, the knowledge of their oxidation characteristics, specifically those related to the (CH_3O–) and (–OCH_2O–) moieties in fuel molecules, can benefit the kinetic understanding of larger POMDMEs oxidation. Considerable research efforts have been devoted to the oxidation chemistry responsible for the low-temperature reactivity of DME [6–12], while the oxidation kinetics of DMM is not as well-established as that of DME. An early

* Corresponding author at: Center for Combustion Energy and Department of Energy and Power Engineering, Tsinghua University, Beijing 100084, PR China.
** Corresponding author at: CNRS-INSIS, I.C.A.R.E.,1C, Avenue de la recherche scientifique, 45071 Orléans cedex 2, France.
E-mail addresses: byang@tsinghua.edu.cn (B. Yang), dagaut@cnrs-orleans.fr (P. Dagaut).

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jet-stirred reactor (JSR) oxidation study [13] for DMM covered medium-to-high temperatures of 800–1200 K, above the typical low-temperature reactivity regime. High-pressure oxidation experiments covering wide ranges of temperatures and equivalence ratios were carried out in a flow tube by Marrodán et al. [14]. Plateaus in fuel concentrations were observed under oxidizing conditions and the authors in [14] proposed a kinetic model incorporating the chemistry of the $CH_3OCH_2O\dot{O}$ radical to account for such phenomena, but chain-branching reactions related to the DMM fuel radicals were not considered. Very recently, during the preparation for this work, Vermeire et al. [15] reported speciation measurements from DMM oxidation in a JSR at the pressure of 1.07 bar and different equivalence ratios. They also proposed a kinetic model which incorporates specific rate coefficients computed at the CBS-QB3 level of theory.

In this work, reactivity and speciation behaviors in DMM oxidation were further pursued via experimental and modeling approaches. Experimental efforts were made by combining advantages of two JSR facilities equipped with different analytical techniques. A DMM/O_2/Ar mixture was investigated in a JSR operated at near-atmospheric (750 Torr) pressure, and the use of a photoionization molecular-beam mass spectrometer (PI-MBMS) enabled the detection of reactive intermediates including peroxides. To be relevant to practical high-pressure conditions in engines, oxidation experiments covering different equivalence ratios (0.2, 0.5 and 1.5) were also performed using another JSR facility at 10 atm, with species concentrations monitored by a Fourier transform infrared spectrometer (FTIR) and gas chromatography (GC). A new kinetic model was proposed to depict DMM oxidation, which was tested with speciation measurements obtained in the current work as well as in literature. The validated kinetic model was used to account for the fuel reactivity and the speciation behavior in different temperature regimes. The consumption behavior of DMM results from the different fates of the two fuel radicals, and similar reaction patterns can be expected for relevant radicals produced during the oxidation of larger POMDMEs.

2. Experimental methods

2.1. Near-atmospheric pressure JSR oxidation

The near-atmospheric pressure JSR experiment was carried out at the Advanced Light Source (ALS) of the Lawrence Berkeley National Laboratory. The apparatus consists of a 33.5 cm^3 spherical reactor, a molecular-beam sampling system and a time-of-flight (TOF) mass spectrometer. Detailed descriptions have been presented earlier [9]. A fuel-lean ($\phi = 0.5$) mixture diluted by argon (Ar) containing 2%DMM/16%O_2 (by mole fractions) was introduced into the JSR, inside which the pressure was kept constant at 750 Torr. The relatively high initial fuel concentration was chosen to result in strong signals of the targeted low-temperature oxidation intermediates. The temperature monitored by a K-type thermocouple was increased from 460 to 820 K with an increment of 20 K. To maintain the residence time of 3 s, flow rates were adjusted accordingly. The burnt gases were sampled via a quartz nozzle, with the sampled species "frozen" by the formation of a molecular-beam, and then analyzed by the photoionization mass spectrometer with a mass resolution ($m/\Delta m$) of ~4000. By scanning the photon energy from 9.6 to 11.2 eV with the temperature controlled at 600 K, photoionization efficiency (PIE) spectra were obtained for species identification [16,17]. At each operating temperature, mass spectra were recorded at photon energies of 16.65, 14.35, 13.20, 11.50, 11.00 and 10.80 eV, to derive the temperature-dependent species mole fractions by following the procedure proposed in our recent work [18]. As summarized in previous publications [19,20], uncertainties of the quantitative measurements obtained with the PI-MBMS technique originate from many aspects: the probe effects, mass discrimination factors, flow controllers, photon flux (calibration of the photodiode), cross sections, and the data evaluation procedures including the identification of fragments. For concentrations of major species (DMM, O_2, Ar, CO_2, CO, H_2O), the measuring error was estimated within ± 20%, and uncertainty factors ranging from 1.3 to 3 could be expected for stable molecular intermediates, mainly depending on the reliability of the adopted photoionization cross section (PICS) values. For reactive intermediates, additional uncertainties can arise from the potential losses during the sampling process. Uncertainty factors for individual species, as well as PICSs and mass discrimination (MD) factors used in the quantification, are provided in Table S1 of the Supplemental material. The thermocouple used to measure the gas temperature was placed at the exit of the reactor near the tip of the sampling probe whose base was water-cooled. So the recorded values should be lower than the actual temperatures because of the probe effects as a heat sink [12]. In previous works [12,18], corrections have been made by linearly shifting the recorded temperatures towards higher values and this practice has been proven reasonable and feasible. For each experiment, the shifting extent mainly depends on the actual location of the thermocouple. After testing the setup with different fuels [9,12,18], the shifting factor was found to range from 5% to 10%. In the current experiment, a value of 6% was chosen to best match the experimental and simulated characteristic temperatures in the fuel profile. The uncertainty of the corrected temperatures was estimated to be within ± 5%.

2.2. High-pressure JSR oxidation

The high-pressure JSR facility in Orléans has been employed in kinetic investigations for diverse fuels [6,21,22], and the experiments' repeatability is within 10%, and also within the total uncertainty of the quantitative data. The fused silica reactor with the volume of 38 cm^3, which is settled inside a pressure-resistant steel chamber, can be operated under high-pressure (10 atm) conditions by balancing the pressures inside and outside the reactor [21]. Highly nitrogen-diluted (1‰ DMM by mole fraction) mixtures were adopted to minimize temperature gradients caused by heat release. Experiments were performed for three mixtures with different equivalence ratios (0.2, 0.5 and 1.5) in the temperature range of 500–1200 K. The liquid fuel was metered with a HPLC pump into a vaporizer while O_2 and N_2 were separately regulated by thermal mass-flow controllers, and the mixed reactants were preheated before entering the reactor. For experiments performed at a given equivalence ratio, at each temperature, flow rates of both the liquid fuel and the gases were altered accordingly to maintain the parameters, including the residence time and the fuel concentration, at constant values (0.7 s and 1‰, respectively). The specifications of the HPLC pump and mass flow controllers allowed to cover the desired flow rates. Exhaust mixtures were sampled with a sonic probe and transferred to an on-line Fourier transform infrared (FTIR) spectrometer and also collected in Pyrex bulbs for further off-line gas chromatography (GC) analyses. The GCs were equipped with different columns for corresponding targeted species (DB-624 for oxygenates, CP-Al_2O_3-KCl for hydrocarbons, and Carboplot-P7 for hydrogen and oxygen), and an electron ionization (70 eV) mass spectrometer was used for species identification. Errors for measured temperatures and concentrations were estimated to be ± 5 K and < ± 15% [22], respectively.

Species mole fraction measurements in all above-mentioned experiments are provided in the Supplemental material.

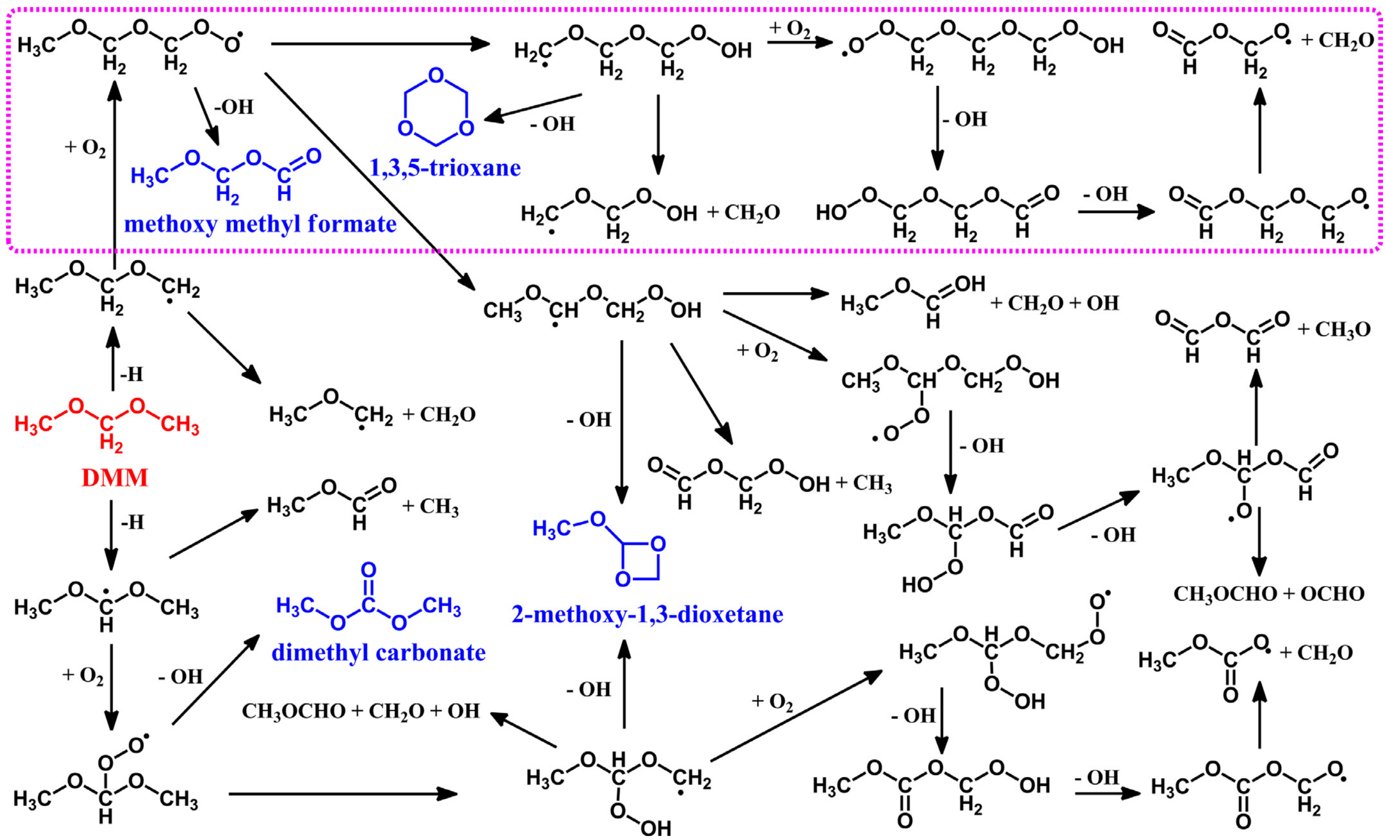

Fig. 1. The schematic for major DMM oxidation pathways considered in the model construction. (Species marked in blue are possible isomers with *m/z* of 90.03 and the framed reactions are responsible for the formation and consumption of $\dot{C}H_2OCH_2OCH_2OOH$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Rate coefficients used in the sub-mechanism for DMM low-temperature oxidation in this work. (Units, cm^3/s/mol/cal).

No.	Reaction		*A*	*n*	E_a	Ref.
R1	$CH_3OCH_2OCH_3 + OH = CH_3OCH_2OCH_2 + H_2O$		1.08×10^3	3.12	-1.73×10^3	[15,26], this work
R2	$CH_3OCH_2OCH_3 + OH = CH_3OCHOCH_3 + H_2O$		5.31×10^8	1.38	3.06×10^2	[15,26], this work
R3	$CH_3OCH_2OCH_3 + CH_3O_2 = CH_3OCH_2OCH_2 + CH_3O_2H$		1.32×10^1	3.56	1.27×10^4	[15], this work
R4	$CH_3OCH_2OCH_3 + CH_3O_2 = CH_3OCHOCH_3 + CH_3O_2H$		2.26×10^2	3.16	1.18×10^4	[15], this work
R5	$CH_3OCH_2OCH_2 + O_2 = COCOCO_2J$	1 atm	4.11×10^{19}	−2.45	1.14×10^3	[11,27]
		10 atm	1.12×10^{14}	−0.58	-1.64×10^2	
R6	$COCOCOOJ = CJOCOCO_2H$		6.70×10^7	0.59	1.41×10^4	[15]
R7	$COCOCOOJ = COCOC^*O + OH$		2.82×10^6	1.97	3.49×10^4	[15]
R8	$CJOCOCO_2H$ = cy-COCOCO + OH		2.51×10^{16}	−1.45	1.56×10^4	[15]
R9	$CJOCOCO_2H = CH_2O + CH_2OCH_2O_2H$	1 atm	4.75×10^{48}	−11.15	3.71×10^4	[28]
		10 atm	2.46×10^{40}	−8.33	3.55×10^4	
R10	$CJOCOCO_2H + O_2 = O_2JCOCOCO_2H$		7.00×10^{11}	0.00	0.00	[8,10,11]
R11	$O_2JCOCOCO_2H = HO_2COCOC^*O + OH$		5.68×10^{11}	−0.04	1.84×10^4	[15]
		Duplicate	1.55×10^2	3.31	3.42×10^4	
R12	$HO_2COCOC^*O = OJCOCOC^*O + OH$		1.50×10^{16}	0.00	4.29×10^4	[15]
R13	$OJCOCOC^*O = CH_2O + OCH_2OCHO$		2.33×10^{13}	0.16	2.34×10^4	[15]
R14	cy-COCOCO = $CH_2O + CH_2O + CH_2O$		1.91×10^{15}	0.00	4.75×10^4	[29]
R15	cy-COCOCO + OH => $H_2O + CH_2O + CH_2O$ + HCO		8.19×10^{12}	0.00	4.60×10^2	[29]
R16	cy-COCOCO + H => $H_2O + CH_2O + CH_2O + H_2$		1.72×10^8	1.90	6.75×10^3	[29]
R17	cy-COCOCO + O => $H_2O + CH_2O + CH_2O$ + OH		5.43×10^{13}	0.00	7.08×10^3	[29]
R18	cy-COCOCO + HO_2 => $H_2O_2 + CH_2O + CH_2O$ + HCO		1.23×10^4	2.50	1.42×10^4	[29]
R19	cy-COCOCO + O_2 => $HO_2 + CH_2O + CH_2O$ + HCO		3.69×10^6	3.00	5.60×10^4	[29]
R20	cy-COCOCO + CH_3 => $CH_4 + CH_2O + CH_2O$ + HCO		1.09×10^{-6}	5.42	5.00×10^3	[29]

3. Kinetic modeling

A new kinetic model for DMM low-temperature oxidation was proposed based on the strategy of "reaction classes" [23–25]. Major pathways considered in the model construction are demonstrated in Fig. 1. Rate coefficients assignments for key reactions will be depicted in detail below, with selected ones listed in Table 1. Names of species included in the DMM oxidation sub-mechanism are provided in Fig. S2.

Reaction sequences shown in Fig. 1 are initiated by hydrogen abstractions from DMM leading to the two fuel radicals, $CH_3OCH_2O\dot{C}H_2$ and $CH_3O\dot{C}HOCH_3$. A high-level theoretical investigation was carried out by Kopp et al. [28] for hydrogen abstractions from DMM by H and CH_3, and the sequential fuel radical

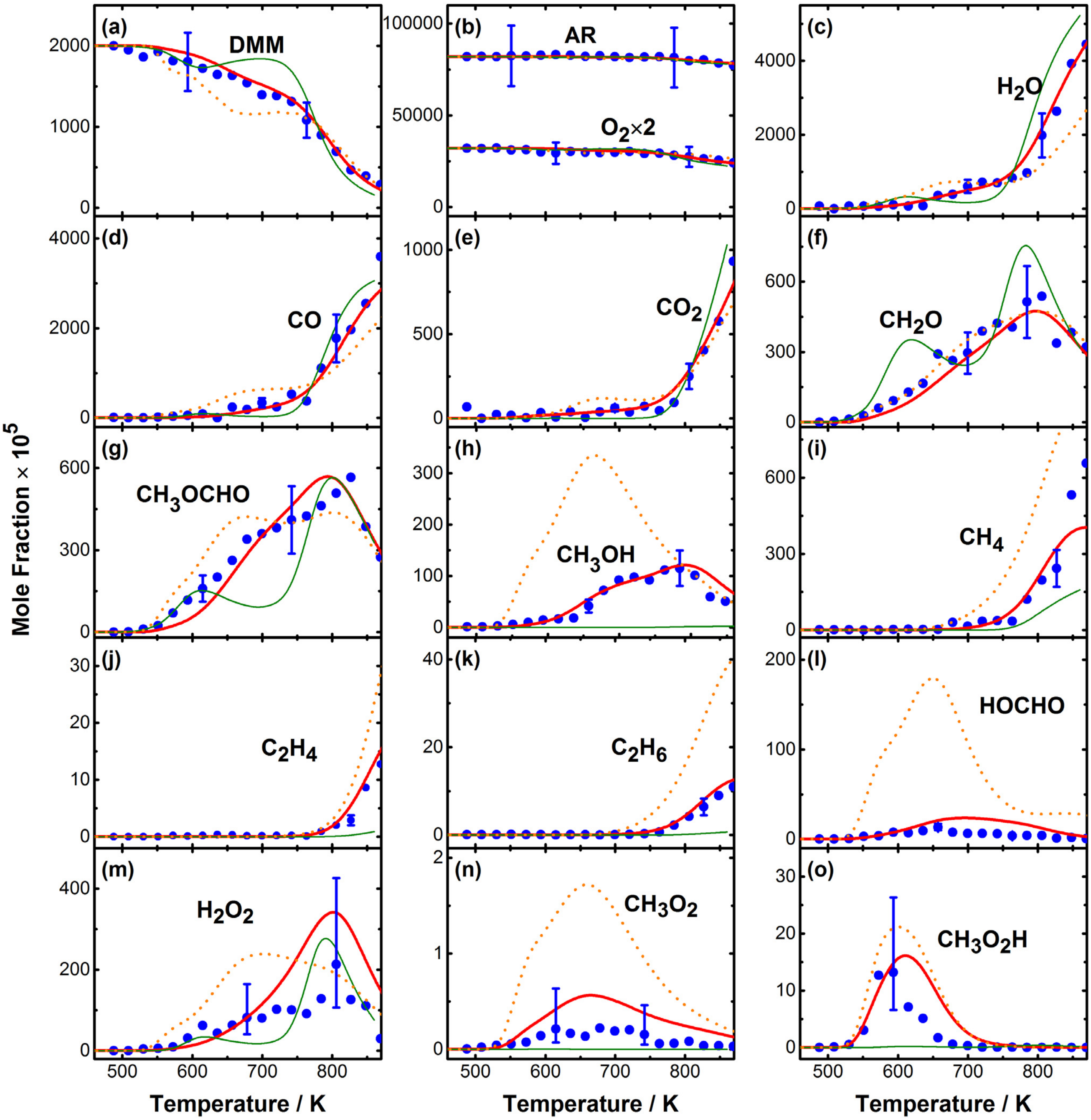

Fig. 2. Mole fraction profiles for species in DMM JSR oxidation at 750 Torr. Symbols, measurements; thick solid lines, simulations with the current model; thick dotted lines, simulations with the Vermeire model; thin solid lines, simulations with the Marrodán model.

consumption. In [28], conventional thermal decompositions as well as hot β-scissions [30] were taken into consideration and the torsional anharmonicity was included in the master equation analysis. The derived temperature- and pressure-dependent rate coefficients which are expected of high accuracy were incorporated in the current model. Furthermore, OH radicals are expected to be a significant abstracting agent in both low- and high-temperature regimes [25]. Overall rate constants for hydrogen abstractions from DMM by OH in previous models [14,15,26,31] are compared in Fig. S3. The results from [26] match the only experimental measurement [32] and therefore were adopted here. The theoretical branching ratio [15] between abstractions from the two different sites was used to compute and fit rate coefficients for separate reactions (R1 and R2 in Table 1). For hydrogen abstractions by O, O_2, CH_3O and HO_2, theoretical rate coefficients provided by Vermeire et al. [15] were employed. Hydrogen abstractions by CH_3O_2 (R3 and R4), which were not considered by Vermeire et al. [15], were introduced into the current model, with the same rate coefficients used for the abstractions by HO_2.

Peroxy (RO_2) radicals, $CH_3OCH_2OCH_2O\dot{O}$ and $CH_3OCH(O\dot{O})OCH_3$, are formed through O_2 additions to corresponding fuel radicals. Intramolecular hydrogen transfers leading to hydroperoxyalkyl (QOOH)-type radicals can proceed through two reactions for $CH_3OCH_2OCH_2O\dot{O}$ and via only one for $CH_3OCH(O\dot{O})OCH_3$. Reactions responsible for the formation and consumption of one QOOH-type radical $\dot{C}H_2OCH_2OCH_2O_2H$, as framed in Fig. 1, are taken as an example to illustrate how kinetic parameters were determined. The first O_2 addition reaction, which is usually temperature- and pressure-dependent, is a crucial step [25] leading to the low-temperature reaction sequence. Usually, variational methods, which are expensive and time-consuming, are required for computing accurate rate coefficients for this barrierless association step [25,33]. For the O_2 addition to $CH_3OCH_2O\dot{C}H_2$, the corresponding DME reaction ($CH_3O\dot{C}H_2 + O_2 \rightleftharpoons CH_3OCH_2O\dot{O}$) serves as the closest available analog based on the structural similarity, and this DME reaction has been extensively investigated through experimental and theoretical approaches. Rate coefficients with different origins [34–36] fall in a three-fold uncertainty around the measurement by Rosado-Reyes et al. [27]. This rate expression, incorporating the pressure-dependence, was used for the target DMM reaction (R5) in the current model. For reaction classes involving tight cyclic transition states, namely RO_2 isomerization (R6, R7), cyclic ether formation from QOOH (R8), and ketohydroperoxide (KHP) formation from O_2QOOH (R11), the theoretical kinetic parameters reported by Vermeire et al.[15] were employed. For the β-scission of $\dot{C}H_2OCH_2OCH_2O_2H$ (R9), the rate constant were assumed to be the same as that of $CH_3OCH_2O\dot{C}H_2$ β-scission [28] with the similar pattern producing CH_2O through the C–O fission. For the secondary O_2 addition, the rate coefficients which were consistently adopted for the corresponding DME reaction in previous models [8,10,11] were used for the similar DMM reaction (R10) in the current model. The six-membered cyclic ether formed from $\dot{C}H_2OCH_2OCH_2O_2H$, 1,3,5-trioxane (cy-COCOCO), is a typical formaldehyde (CH_2O) precursor, whose gas-phase kinetics has been investigated in previous works [29,37,38]. The sub-mechanism responsible for its consumption (R14–R20) was taken from a recent kinetic model proposed by Santner et al. [29].

For reactions related to the other two QOOH-type radicals, $CH_3O\dot{C}HOCH_2OOH$ and $CH_3OCH(OOH)O\dot{C}H_2$, the similar procedure was followed to determine the corresponding rate coefficients. The AramcoMech 2.0 [39] which contains the oxidation chemistry of relevant smaller intermediates was adopted as the core mechanism for the current DMM model. Theoretically computed thermodynamic properties for species including the fuel, fuel radicals, RO_2 and QOOH radicals by Kopp et al. [28] were adopted. For other fuel-related species, THERM [40] was employed to calculate the thermochemical data with the latest group values reported in [41]. The entire model involving 524 species and 2821 reactions is provided in the Supplemental material. The perfectly stirred reactor (PSR) module in the ChemkinPro [42] package was used for simulations in this work. Time-independent solutions were obtained using the transient solver with the end time set at 100 s. To point out, the experimental inlet flow rates and the corrected temperatures were used to simulate the measurements obtained at 750 Torr. This leads to the residence time ranging from 2.8 to 3.0 s.

4. Results and discussion

4.1. Near-atmospheric pressure JSR oxidation

About a dozen of products and intermediates (see Table S1) were detected from the DMM oxidation at 750 Torr, including some reactive species such as hydroperoxide (H_2O_2), methyl peroxy (CH_3O_2) and methyl hydroperoxide (CH_3O_2H). However, only a few intermediates observed in the DME oxidation [9] with the same JSR facility and PI-MBMS technique were present in the current DMM oxidation species pool. The common (CH_3OCH_2–) moiety in both fuels is expected to yield many common intermediates. The relatively scanty species pool indicates that DMM is less reactive, which is quite noteworthy because compared to DME, DMM has a longer chain enabling a potentially greater chance of reactivity initiated by hydrogen abstractions and occurrences of intramolecular hydrogen shifts through cyclic transition states.

Measured mole fraction profiles for detected species are presented in Fig. 2, together with predictions by the current model as well as the recent models by Vermeire et al. [15] and Marrodán et al. [14] as references. Improvements in the predictive performances can be seen for the current model, including two aspects: Specific intermediates missing from the Marrodán model, like formic acid (HOCHO), can be predicted; Shapes and magnitudes of species mole fraction profiles can be better reproduced. Regarding the fuel conversion behavior, the measured three-stage fuel decay can be well captured by the current model while the Marrodán model predicts a negative temperature coefficient (NTC) zone in the range of 600–700 K and the Vermeire model over-estimates the fuel reactivity at temperatures from 600 to 750 K. The stepwise formation

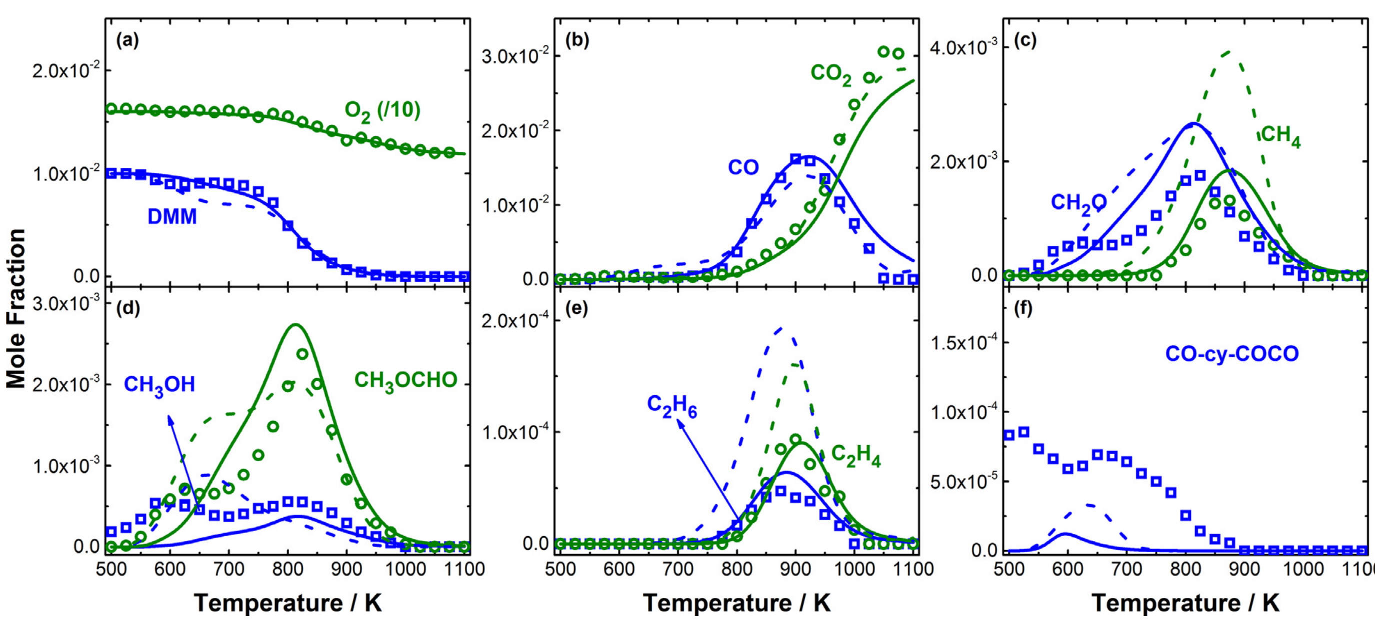

Fig. 3. Model validations against species mole fractions measured by Vermeire et al. [15] during the JSR oxidation of DMM at $\phi = 0.25$, $p = 1.07$ bar, $\tau = 2.83$ s. Symbol, measurements; Solid lines, simulations with the current model; Dashed lines, simulations with the Vermeire model.

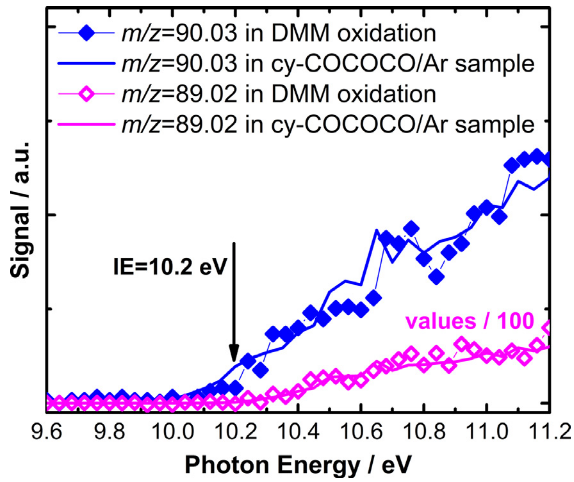

Fig. 4. The PIE curves for peaks with m/z of 89.02 and 90.03 recorded, respectively, in DMM oxidation and a gaseous cy-COCOCO/Ar sample.

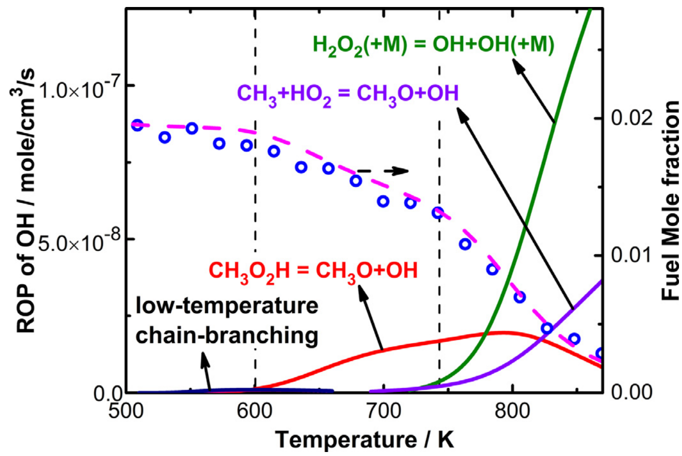

Fig. 5. ROP analysis for OH radical (solid lines to the left axis) and experimental and simulated fuel mole fractions (symbols and the dashed line to the right axis) at different temperatures.

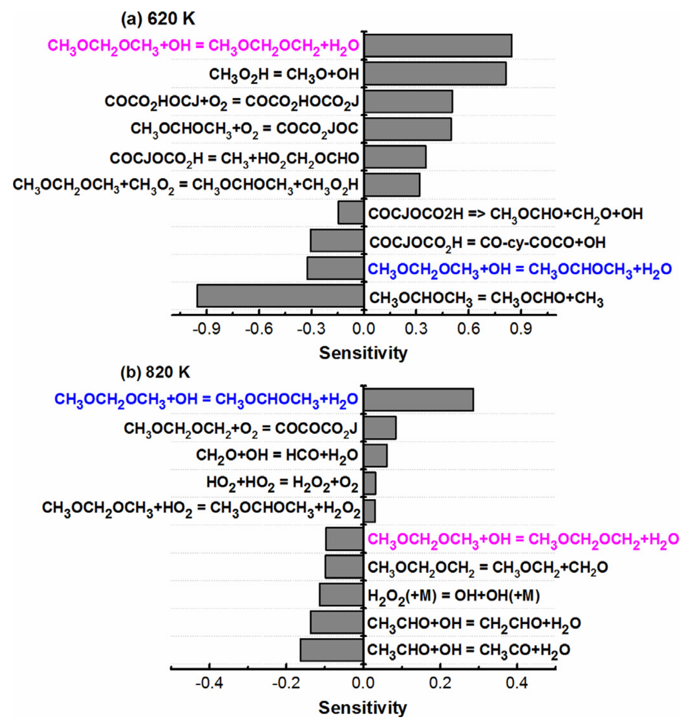

Fig. 6. Sensitivity analyses for CH_3OCHO concentrations at (a) 620 K and (b) 820 K.

pattern of H_2O which is model-predicted to be produced through hydrogen abstractions by OH from the fuel, is in line with the fuel consumption. This indicates the role of OH radicals as preponderant chain carriers in the entire investigated temperature range. The fuel's carbon atoms eventually end up in CO and CO_2 mainly via formaldehyde (CH_2O) and methyl formate (CH_3OCHO). Mole fraction distributions of these two main intermediates can be satisfactorily captured by both the current model and the Vermeire model, when experimental and modeling uncertainties are taken into consideration. Vermeire et al. [15] mentioned the over-prediction for hydrocarbon concentrations as a to-be-improved aspect of their kinetic model, while this issue is not observed with the current model. CH_3O_2 and CH_3O_2H, particularly the former, are reactive intermediates with potential losses on the sampling cone; their measured concentrations are then typically lower than actual values. Simulations by the Marrodán model are however magnitudes lower than the measurements. The Vermeire model gives reasonable predictions, but the current model demonstrates a better performance, which is mainly because CH_3O_2 has been considered as one of the hydrogen abstracting agents in the model proposed in this work.

The predictive performance of the current model was also evaluated against the recent measurements by Vermeire *et al.* [15] at the similar pressure of 1.07 bar. Quantitative species profiles measured by GC and the corresponding simulations at the equivalence ratio of 0.25 are shown in Fig. 3 for comparison. Those under more fuel-rich conditions ($\phi = 1.0$ and 2.0) are provided in the Supplemental material. Both models can satisfactorily capture the mole fraction distributions of the presented species, while the current model better depicts the fuel reactivity and improves the predictions for hydrocarbon intermediates, in line with the observations from Fig. 2, as discussed above.

Vermeire et al. [15] reported the detection of a cyclic ether species, methoxy–1,3-dioxetane (CO-cy-COCO) in their experiments, and the mole fraction profiles were also provided [see Fig. 3(f)]. But it seems the signal was perturbed as CO-cy-COCO already exists before the fuel consumption starts. Also, CO-cy-COCO has similar measured mole fractions at different equivalence ratios, which is quite unexpected because concentrations of such a typical low-temperature oxidation product should be sensitive to the stoichiometry.

Nevertheless, the presence of this four-membered cyclic ether species could not be evidenced by the current experiments employing the PI-MBMS technique. In our work, weak signals from peaks with m/z of 89.02 and 90.03, respectively, belong to $C_3H_5O_3$ and $C_3H_6O_3$ species, according to their molecular weights measured by the high-mass-resolution mass spectrometer. The signal of $C_3H_6O_3$ may come from four possible isomers shown in the schematic pathways (see Fig. 1), i.e. carbonyl compounds including dimethyl carbonate and methoxy methyl formate and cyclic ethers including 1,3,5-trioxane (cy-COCOCO) and CO-cy-COCO. Adiabatic ionization energies were computed at CBS-QB3 and M06-2X/aug-cc-pvtz levels of theory for these candidates, as provided in Table S2. An ionization onset appears around 10.2 eV in the measured PIE curve for $C_3H_6O_3$ ($m/z = 90.03$) in Fig. 4. However, calculated IEs for all above-mentioned isomers except dimethyl carbonate are close to this value given the calculating error of ± 0.2 eV [9]. According to the potential energy surface (PES) theoretically derived in [15], pathways leading to carbonyl intermediates have relatively high energy barriers and thus are not competitive. Therefore, the two cyclic ethers are more likely candidates. An argon-diluted cy-COCOCO sample was prepared for a photon energy scan, in which a peak with m/z of 89.02 was found to come from the fragmentation of cy-COCOCO. PIE curves for both $m/z = 90.03$ and $m/z = 89.02$ peaks probed from the DMM oxidation were compared with those in the reference mixture. The good match in shapes, as can be seen from Fig. 4, indicates a great likelihood of the presence of cy-COCOCO, and thus its precursor, a QOOH-type radical $\dot{C}H_2OCH_2OCH_2OOH$, in DMM oxidation.

To unravel the chemistry governing the reactivity of DMM, ROP analysis was performed for OH, the crucial chain carrier, with the temperature-dependent results shown in Fig. 5. At temperatures below 600 K, where the fuel has negligible reactivity, low-temperature chain-branching reaction sequences, mainly the consumption of O_2QOOH produced from 2nd O_2 additions, only provide a small amount of OH. Such reactions are of kinetic significance for fuels with obvious low-temperature reactivity like DME, as consistently pointed out in previous investigations [9–12]. If tracing back the fuel consumption pathways in this temperature region, at 570 K for example, a branching ratio of around 2:3 is model-predicted between hydrogen abstractions respectively leading to fuel radicals $CH_3OCH_2O\dot{C}H_2$ and $CH_3O\dot{C}HOCH_3$. The former almost exclusively goes through O_2 addition, but about 50% of the dominant $CH_3O\dot{C}HOCH_3$ rapidly decomposes through β-scission yielding CH_3 and the relatively stable CH_3OCHO. Besides, the dominant QOOH-type radical, $CH_3O\dot{C}HOCH_2OOH$, behaves similarly to $CH_3O\dot{C}HOCH_3$. It is predominantly consumed through β-scissions and the formation of cyclic ether, while the 2nd O_2 addition has a limited contribution (less than 10% at 570 K).

Fuel is consumed faster when increasing the temperature above 600 K, as the dissociation of CH_3O_2H ($CH_3O_2H \rightleftharpoons CH_3O + OH$) yields OH. It can be seen from Fig. 2(o) that the peak concentration of CH_3O_2H was measured to be 130 ppm, which is more than one order of magnitude higher than that in the DME oxidation under more fuel-lean conditions ($\phi = 0.35$) with a higher fuel concentration (2.5% by mole fraction) [9]. This high peak concentration results from the significant β-scission of $CH_3O\dot{C}HOCH_3$ easily yielding CH_3 which almost exclusively combines with O_2 at low-to-medium temperatures. The resulting CH_3O_2 radicals subsequently convert to CH_3O_2H mainly through $CH_3O_2 + HO_2 \rightleftharpoons CH_3O_2H + O_2$. In the temperature region of 600-740 K, besides the dominant role of CH_3O_2H decomposition, the consumption of $CH_3O\dot{C}HOCH_2OOH$ radical also slightly contributes to OH production through two channels, i.e., the formation of cyclic ether and the β-scission (see Fig. 1). The aforementioned reactions result in the mild fuel consumption till OH is massively produced from H_2O_2 decomposition over 740 K. Then, the fuel decays dramatically and hydrocarbon intermediates such as CH_4 and C_2H_6 start to accumulate, marking the transition to the high-temperature oxidation. The reaction $CH_3 + HO_2 \rightleftharpoons CH_3O + OH$ has a non-negligible contribution to OH production at higher temperatures (> 740 K). This further indicates the importance of CH_3 chemistry to DMM oxidation because of the prevailing β-scission of $CH_3O\dot{C}HOCH_3$. Over the entire investigated temperature range, this reaction con-

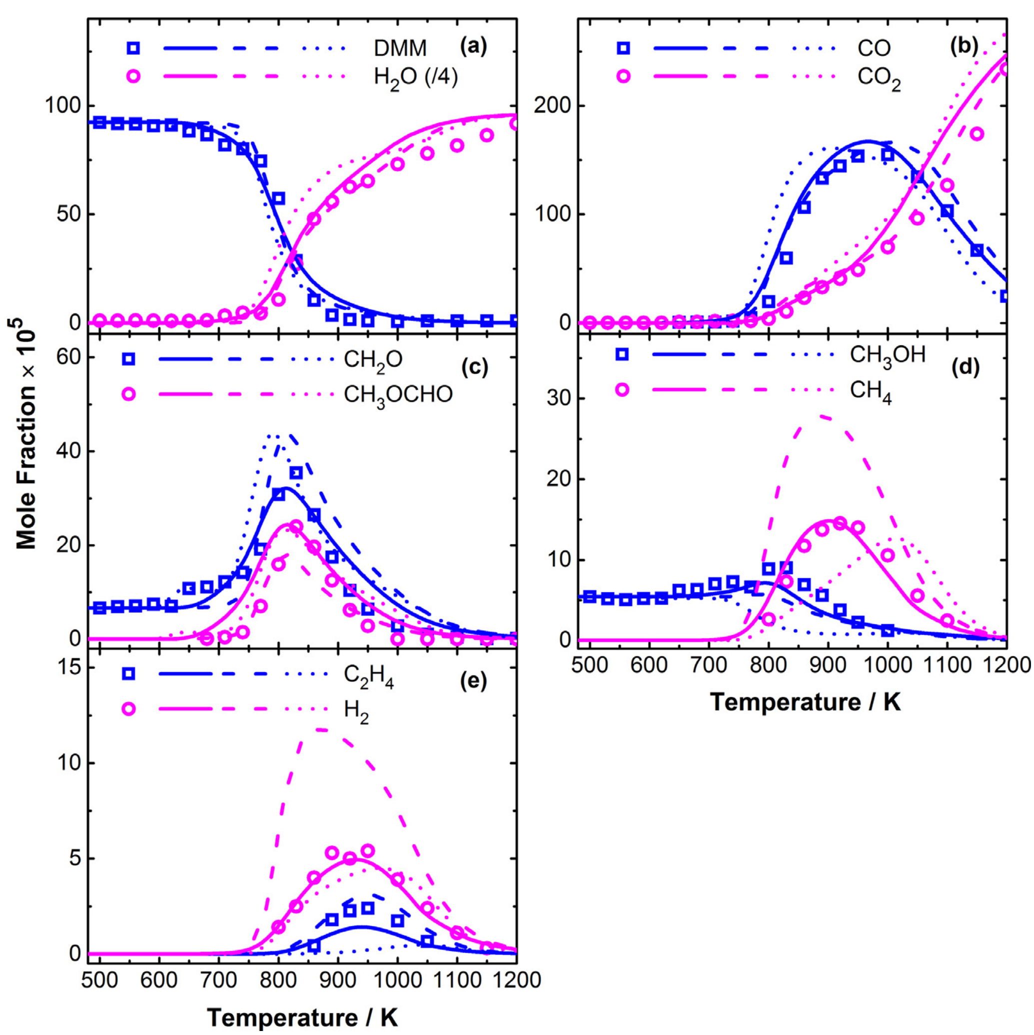

Fig. 7. Experimental (symbols) and simulated (lines) species concentrations in DMM JSR oxidation at $\phi = 0.2$, $p = 10$ atm, $\tau = 0.7$ s. Solid lines, simulations with the current model; dashed lines, simulations with the Vermeire model dotted lines, simulations with the Marrodán model.

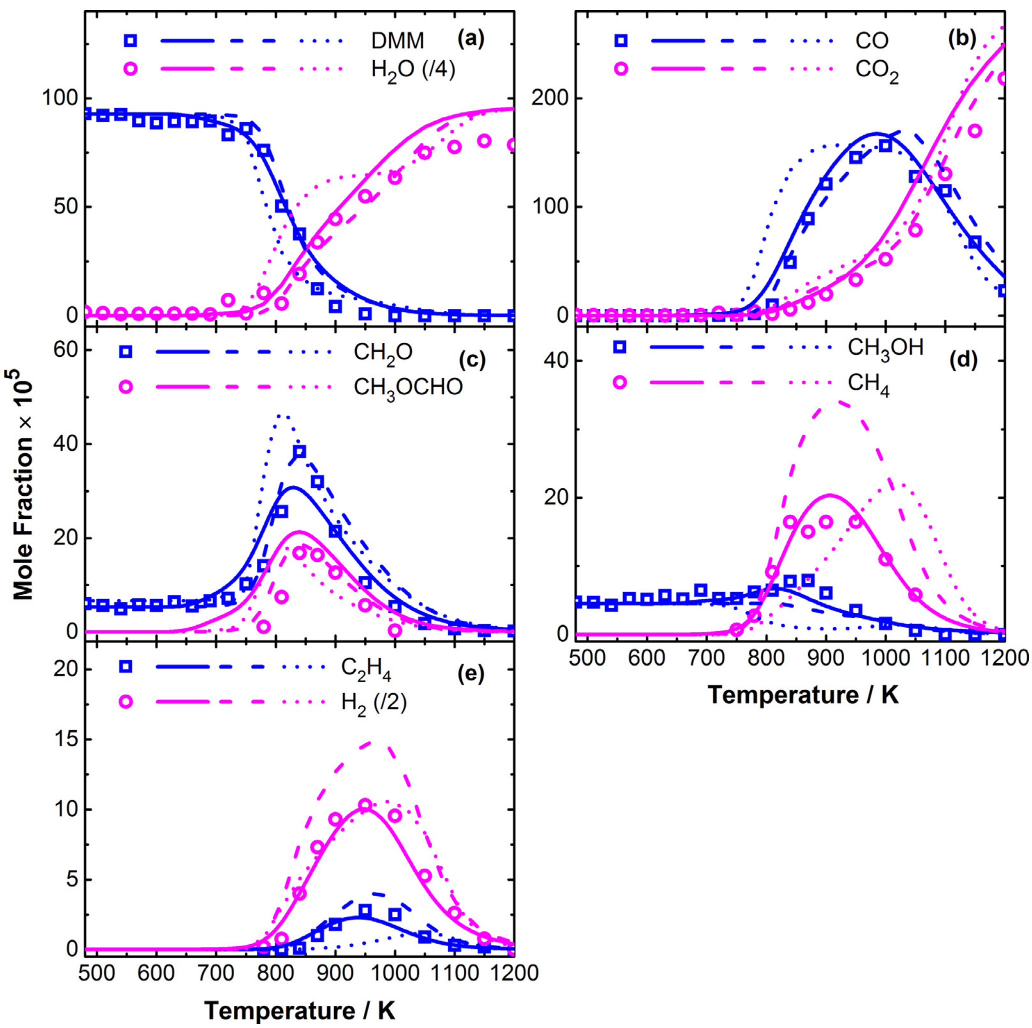

Fig. 8. Experimental (symbols) and simulated (lines) species concentrations in DMM JSR oxidation at $\phi = 0.5$, $p = 10$ atm, $\tau = 0.7$ s. Solid lines, simulations with the current model; dashed lines, simulations with the Vermeire model; dotted lines, simulations with the Marrodán model.

tributes over 90% to the formation of another crucial intermediate CH_3OCHO, for which sensitivity analyses at 620 K and 820 K are displayed in Fig. 6. The competing pair of hydrogen abstractions by OH from different sites in DMM have opposite effects on CH_3OCHO concentrations. The reaction generating $CH_3O\dot{C}HOCH_3$ unexpectedly inhibits CH_3OCHO formation at 620 K, because at this temperature the radical (mainly OH) level is relatively low, limiting the fuel consumption and intermediates formation. Thus, the production of $CH_3OCH_2O\dot{C}H_2$, which can result in chain-branching as mentioned above, shows positive sensitivity. At higher temperatures the fuel consumption is no longer limited by OH concentrations, so under such conditions CH_3OCHO formation is promoted by the production of its precursor $CH_3O\dot{C}HOCH_3$. The current model can well predict CH_3OCHO concentrations at different temperatures, indicating that the adopted branching ratio between hydrogen abstractions from different sites of DMM is reasonable.

4.2. High pressure JSR oxidation

In the high-pressure oxidation experiments of DMM, the types of species measured are within the scope discussed above. Figures 7–9 display speciation measurements at three different equivalence ratios. The initial fuel concentrations were measured to be around 9.2×10^{-4}, indicating the occurrences of slight fuel decomposition attributed to catalytic reactions in the vaporizer. To consider such influences and to account for the initial platforms in CH_2O and CH_3OH concentrations (see Figs. 7–9), the measured mole fractions of DMM, CH_2O and CH_3OH at the lowest temperatures were taken as the input to perform the simulations. The validity of such corrections is detailed in the Supplemental material. Simulated results with the current and the literature [14,15] models in literature are shown in Figs. 7–9 for comparison. The problem of over-predictions for hydrocarbons of the Vermeire model still exists under high pressure conditions. The Marrodán model cannot capture the fuel reactivity well, particularly under fuel-rich conditions. While the current model shows satisfactory performances under all investigated conditions.

Negligible low-temperature reactivity was observed during DMM high-pressure oxidation even in the very fuel-lean ($\phi = 0.2$) case. Under high-pressure conditions, the dominant fuel radical $CH_3O\dot{C}HOCH_3$ has even higher tendency to decompose, because at investigated temperatures the rate coefficient at 10 atm is several times larger than that at 1 atm, according to the recent theoretical work by Kopp et al. [28]. The consumption of the other fuel radical $CH_3OCH_2O\dot{C}H_2$ is still dominated by O_2 addition and the subsequent isomerization at low temperatures (600–700 K). But at high pressures, the dominant resulting QOOH-type radical, $CH_3O\dot{C}HOCH_2OOH$, tends to decompose through β-scissions, rather than the competing channels of the second O_2 addition or the cyclic ether formation. This is in line with the fact that the cyclic ether species were not observed in the high-pressure experiments.

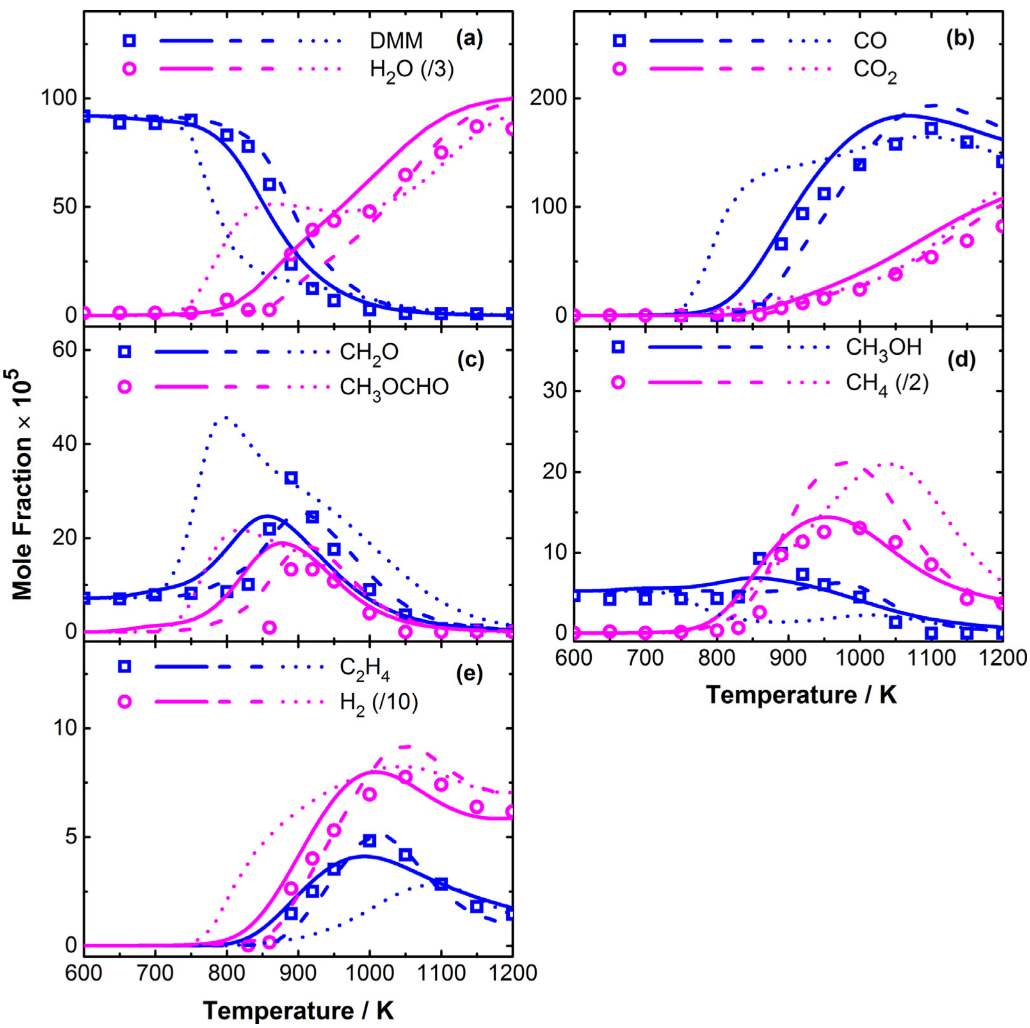

Fig. 9. Experimental (symbols) and simulated (lines) species concentrations in DMM JSR oxidation at $\phi = 1.5$, $p = 10$ atm, $\tau = 0.7$ s. Solid lines, simulations with the current model; dashed lines, simulations with the Vermeire model; dotted lines, simulations with the Marrodán model.

According to the model prediction, the peak concentration of cyclic ethers is at the order of magnitude of 10^{-8}, below the current detection limit of 10^{-6}.

The effects of the equivalence ratio on the fuel consumption pathways were explored by performing ROP analyses at 700 K, with the results shown in Fig. 10.

When increasing the equivalence ratio, a notable difference occurs to the branching ratio between the O_2 addition reaction and the β-scission of the fuel radical $CH_3OCH_2O\dot{C}H_2$, as the relative importance of the low-temperature reaction sequence starting from O_2 addition is further lessened. This not only results in the reduced fuel reactivity, but also changes the formation pathways of the two key intermediates, CH_2O and CH_3OCHO, whose main production channels are summarized in Table S4, based on the ROP-analyzed results. Two products from the low-temperature reactions, $CH_3O\dot{C}HOCH_2OOH$ and CO-cy-COCO, can decompose and release CH_2O and CH_3OCHO simultaneously. These two reactions are important contributors to CH_2O and CH_3OCHO formations at low temperatures, but their significance is weakened when increasing the equivalence ratio (see Table S4). Though the branching ratio between hydrogen abstractions from the different sites of DMM remains almost the same at various equivalence ratios, the relative importance of different abstracting species changes. Under fuel-lean conditions ($\phi = 0.2$ and 0.5), hydrogen abstractions by HO_2 and CH_3O_2 play a significant role. While under fuel-rich conditions ($\phi = 1.5$), CH_3O becomes the second most important abstracting agent after OH. This leads to the dominant CH_3OH formation pathway shifting from the reaction $CH_3O + HO_2 \rightleftharpoons CH_3OH + O_2$ to the hydrogen abstractions from DMM by CH_3O (see Table S4).

Based on above experimental evidences and modeling interpretations, the following reaction scheme accounts for DMM consumption in the low-temperature regime: Hydrogen abstractions from the fuel mainly occur in the central (–OCH_2O–) moiety, and the resulting fuel radical $CH_3O\dot{C}HOCH_3$ swiftly decomposes, impeding the O_2 addition pathway leading to the low-temperature reactivity. Chain-branching can occur via the other fuel radical $CH_3OCH_2O\dot{C}H_2$ which is only produced in a small amount. Distinct consumption behaviors of the two types of radicals can also be used to predict the reactivity of larger POMDME compounds. When increasing the number of (–CH_2O–) units, the possibility of hydrogen abstractions from the (CH_3O–) groups is further reduced. For major fuel radicals with the structure of $CH_3O(CH_2O)_o\dot{C}HO(CH_2O)_pCH_3$ ($o + p \geq 1$, $o \geq 0$, $p \geq 0$), quick β-scissions will dominate over O_2 additions. However, a radical with the structure of $CH_3(OCH_2)_q$ ($q \geq 1$) can form through the β-scission besides a stable carbonyl molecule. Like the investigated $CH_3O\dot{C}H_2$ [10,11] and $CH_3OCH_2O\dot{C}H_2$ [28], such radicals have some thermal stability at low temperatures, thus enabling O_2 additions. Moreover, their decomposition products, $CH_3(OCH_2)_{q-1}$ (when $q \geq 2$) radicals can be expected to show similar kinetic be-

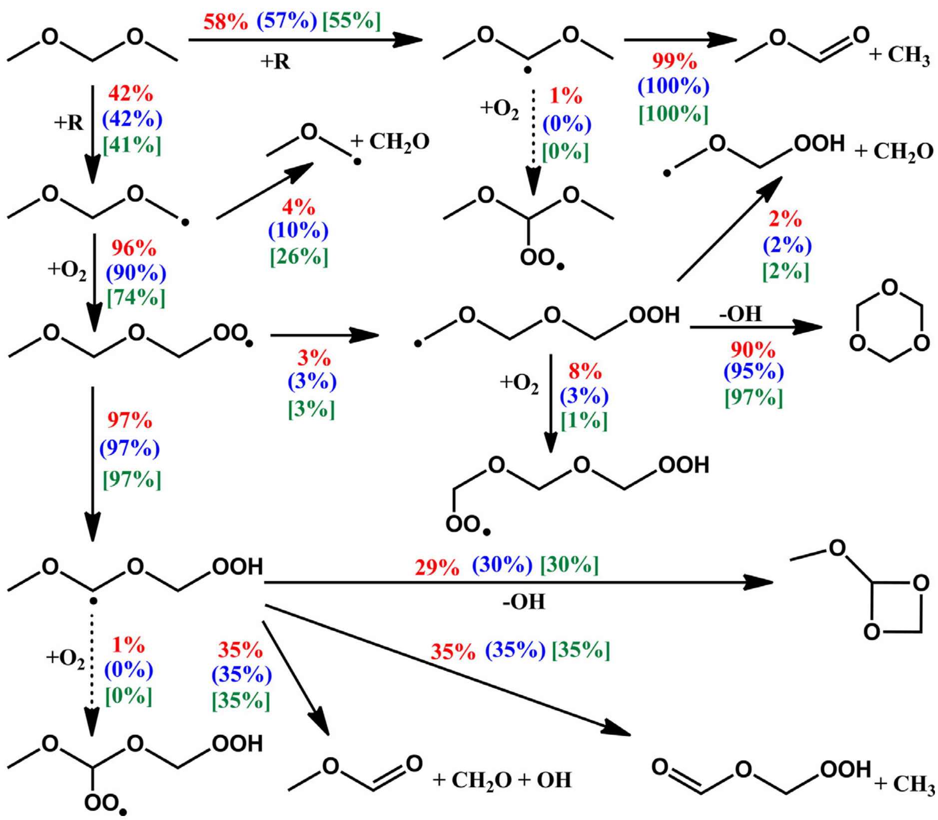

Fig. 10. Major fuel consumption pathways in DMM JSR oxidation at 700 K and 10 atm. The numbers are percent contributions to the consumption of the species on the source side of the arrow at the equivalence ratios of 0.2, 0.5 (in parentheses) and 1.5 [in square brackets].

haviors. In this way, the low-temperature reactivity increases when lengthening the (–CH_2O–) chain for POMDME compounds.

The kinetic model for DMM developed in the current work has shown satisfactory performances under low-temperature oxidation conditions. To further test the current model at higher temperatures, the model was also used to simulate the chemical structure of a laminar premixed flame fueled by DMM reported in our previous work [43]. As demonstrated in Figs. S7 and S8, a good agreement can be seen between the experimental and modeling species profiles.

5. Conclusions

Low-temperature oxidation kinetics of DMM was explored in this work with the combination of experimental and modeling efforts. Speciation information was probed from DMM oxidation in two JSR facilities operated at different pressures. A new kinetic model was proposed based on the "reaction classes" strategy, which can well capture the fuel consumption and intermediates formation behaviors measured under all investigated conditions. The fuel reactivity was found to be governed by two competitive reaction pairs, the hydrogen abstractions from different sites and the O_2 addition versus the β-scission of the dominant fuel radical. The major carbon flux through $CH_3OCH_2OCH_3 \rightarrow CH_3O\dot{C}HOCH_3 \rightarrow CH_3OCHO + CH_3$ hinders the O_2 addition to $CH_3O\dot{C}HOCH_3$ and subsequent chain-branching processes, accounting for the weak reactivity of DMM at low temperatures. Nevertheless, the minor fuel radical $CH_3OCH_2O\dot{C}H_2$ can lead to chain-branching, as the occurrence of $RO_2 \rightleftharpoons$ QOOH isomerization can be proven by the detection of a cyclic ether species, a product from the consumption of a QOOH-type radical. Also, because of the predominant $CH_3O\dot{C}HOCH_3$ β-scission, the CH_3 chemistry is crucial for describing DMM oxidation. The decomposition of CH_3O_2H ($CH_3O_2H \rightleftharpoons CH_3O + OH$) gives rise to a slow fuel consumption before the tremendous fuel destruction initiated by $H_2O_2(+M) \rightleftharpoons OH + OH(+M)$. Kinetic perspectives, such as the consumption patterns of relevant fuel radicals, can also help to understand the oxidation chemistry of larger practical POMDME compounds.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2018.04.008.

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